Forecasting Football Results Using Machine Learning Techniques

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***Abstract*—Football, with its global fanbase of over 3.5 billion, presents a complex system where predicting match outcomes remains challenging because of its inherent unpredictability. This study employs a mixed-methods approach, combining quantitative machine learning (ML) models and qualitative analysis to predict potential wins in football matches. Using Kaggle's European Soccer Database—which contains extensive data on hundreds of matches, players, coaches, and teams across the 2023–2024 seasons of the five top European leagues—we implemented supervised machine learning algorithms, including Random Forest (RF), AdaBoost, and XGBoost, alongside in-depth expert qualitative insights and comprehensive, rigorous, detailed tactical analysis. Quantitative results show that the RF model achieves the highest performance with an accuracy of 68.4% and an F1-score of 0.65, outperforming the baseline betting odds (F1-score: 0.39). Qualitative findings highlight the role of intangible factors, such as team morale and managerial decisions. This study underscores the synergy of quantitative and qualitative methods for robust predictions and provides a replicable framework for sports analytics by integrating ML models with expert knowledge.**

***Index Terms*—Football match prediction, machine learning, Mixed-Methods, European Soccer Database, Kaggle, Random Forest, Logistic Regression, XGBoost.**

I. Introduction

Football, a sport with a global fanbase exceeding 3.5 billion, is a dynamic and complex system in which predicting match outcomes remains a formidable challenge [1]. The inherent unpredictability of the game, driven by factors such as player performance, tactical decisions, and intangible elements such as team morale, has long captivated researchers, analysts, and fans alike. While traditional approaches to prediction, such as statistical models and expert opinions, have provided valuable insights, they often struggle to capture the full spectrum of variables influencing the match results. Recent advancements in machine learning (ML) offer a promising avenue for enhancing predictive accuracy by leveraging large datasets and sophisticated algorithms. However, quantitative models alone may overlook critical qualitative factors, such as managerial strategies or team dynamics, which are difficult to quantify but profoundly impact the outcomes.

Over the past few years, the adoption of machine learning (ML) methods has fundamentally transformed outcome prediction in sports, offering improved accuracy and interpretability through pattern recognition and data-driven insights [2]. Examples of supervised ML algorithms include Logistic Regression (LR), Random Forest (RF), and X GBoost, which have shown promise in classification tasks, including win-loss predictions and player performance modelling [3]. Despite these advances, most existing studies adopt a purely quantitative approach, overlooking qualitative factors such as tactical decisions and psychological momentum, which are often critical in determining match outcomes [4].

This study proposes a mixed-methods approach that combines supervised ML with qualitative insights to predict the outcomes of football matches. Utilizing the European Soccer Database from Kaggle [5], which includes data on variables to support forecasting from the European Top 5 Leagues between 2023 and 2024, we integrated engineered match features (e.g., team strength, recent form, and goal margins) with qualitative themes derived from expert interviews and tactical match reports. By employing this hybrid approach, we aimed to enhance both the predictive performance and model interpretability.

Recent studies have emphasized the importance of feature engineering in enhancing model accuracy, particularly in domains such as football, where contextual and temporal variables significantly affect outcomes [6]. Features such as recent team form, player ratings, home advantage, and betting odds are often used as predictors; however, they are limited in capturing the dynamic nature of in-game events and team psychology [7]. Moreover, ML models trained on historical datasets often face challenges in generalizability, as changes in squad composition, managerial tactics, and league dynamics can lead to data drift over time. These limitations highlight the need for hybrid approaches that integrate domain-specific qualitative data with quantitative features to offer more robust and context-aware predictions. By incorporating expert interviews and match report analyses, this study seeks to mitigate the blind spots of purely data-driven models and offer a richer explanatory power in predictive tasks. Such mixed-methods strategies are increasingly being recognized in sports analytics for their potential to bridge algorithmic precision and human insight [8].

II. Literature Review

A. Football Match Prediction

Football, widely recognized as the world's preeminent sport, owes its universal appeal to its accessibility and its simplicity. Requiring minimal equipment and adaptable to diverse environments— indoors or outdoors, across varied terrains—football transcends geographical and socioeconomic boundaries. Its straightforward rules facilitate participation across genders and age groups, fostering widespread engagement in both recreational and competitive contexts globally. This pervasive popularity has propelled professional football into a formidable economic force characterized by extensive markets encompassing matches, tournaments, and associated industries. Central to the sport's allure is the anticipation surrounding match outcomes and the strategic interventions that may influence them in favor of a preferred team. Consequently, the ability to predict football match results has emerged as a valuable tool, offering insights for stakeholders ranging from coaches and analysts to fans and industry professionals [9]. This study delves into the application of predictive analytics, particularly through machine learning, to forecast match outcomes, thereby enhancing strategic decision-making within the dynamic landscape of football [10].

B. Machine Learning

Machine learning (ML) is revolutionizing epidemiology by leveraging the power of large and complex datasets, commonly referred to as "big data." Traditional statistical methods often fall short in managing the scale and intricacy of such data, whereas ML provides innovative approaches for predictive modeling, data clustering and pattern recognition. Recent research underscores ML's transformative potential of ML in health data analysis, offering enhanced accuracy and insights into disease dynamics [11]. Despite its promise, the integration of ML into epidemiology presents significant challenges, notably the divergence in technical terminology and methodologies between the two fields. This discrepancy hinders epidemiologists' ability to effectively interpret and evaluate ML-based studies. This article explores the application of ML in epidemiology, highlighting its opportunities for advancing public health research while addressing the barriers to its adoption, particularly the need for interdisciplinary collaboration to bridge the gap between ML and epidemiological frameworks [12].

C. Kaggle

Kaggle, launched in 2010, has emerged as a pivotal platform in the fields of data science and machine learning, fostering a global community of practitioners, researchers and enthusiasts. It is often described as a "home for data science." Kaggle provides a collaborative environment in which individuals and teams can engage in competitions, share datasets, and develop machine learning models. This literature review synthesizes existing knowledge about Kaggle, exploring its functionalities, impact on data science education and practice, and role in advancing research and industry applications. Kaggle offers a repository of publicly available datasets covering domains such as healthcare, finance, and social sciences. Users can upload, share, and explore datasets to foster open data initiatives. This feature supports research and experimentation by providing access to diverse and high-quality data [13].

D. Random Forest

Random Forest is an ensemble algorithm that combines multiple decision trees to classify data or predict outcomes, and is suitable and effective for both classification and regression tasks. It generates results by averaging the outputs of individual decision trees, each with a root node and leaf nodes that determine the final prediction results. A key advantage of Random Forest as a classifier is its ability to handle missing data. In this study, the Random Forest method was applied using Python, specifically through the sklearn.ensemble library [14].

E. XGBoost

XGBoost, an acronym for eXtreme Gradient Boosting, is an advanced machine learning technique lauded for its speed and effectiveness in regression, classification, and ranking tasks. It builds trees in sequence, with each successive tree focused on correcting the errors of its predecessors, thereby refining the predictions through gradient-based optimization. The "extreme" part highlights its speed and accuracy, achieved through features like parallel computing, tree pruning, and smart handling of missing data, making it more effective than standard gradient boosting methods [15].

III. Materials and Methods

A. Dataset

The dataset for this study was obtained from Kaggle and processed to determine the results of the algorithms. The dataset obtained from kaggle. com/datasets/kamrangayibov has been released since May 6, 2025, with 10 attributes and one label class to predict football results. The table below shows the processed attributes and their conversion into structured explanations.

B. Methods

1) Data Collections
   Data Football-Data.org. We also added factors such as home or away games and weather conditions. The final dataset included 12,500 matches with over 200 features each, which were cleaned to remove any missing or inconsistent data.
2) Data Preprocessing
   The raw dataset underwent meticulous preprocessing to ensure data integrity and suitability for machine-learning analysis. A systematic cleaning procedure was implemented to address missing values and inconsistencies. To handle missing values, numerical features were filled using the median to maintain the central value of the data, whereas categorical features were completed using the mode to represent the most common category.

Outliers in continuous variables, such as anomalously high shot counts, were identified using the interquartile range (IQR) method and constrained to the 95th percentile to mitigate their influence on model performance. Categorical variables, including team names and match venues, were transformed using one-hot encoding to facilitate their integration into the predictive models. To ensure uniformity across features, numerical variables were normalized using min-max scaling, rescaling them to a common range. To improve efficiency and mitigate overfitting, we applied Recursive Feature Elimination (RFE) with a Random Forest classifier, methodically selecting the top 20 most influential features for use in our subsequent analyses.

3) Model Development
To estimate football match results—classified as win, draw, or loss—this study applied three different machine learning techniques: Random Forest (RF), eXtreme Gradient Boosting (XGBoost), and Adaptive Boosting (AdaBoost). The RF model was built using 100 decision trees, with a maximum depth set to 10. A grid search method was used to fine-tune the hyperparameters for better prediction accuracy. For the XGBoost model, a learning rate of 0.1 and 150 estimators were chosen, and early stopping was added to limit the overfitting. The Deep Neural Network (DNN) consisted of three hidden layers with 128, 64, and 32 neurons. ReLU activation functions were used in the hidden layers, whereas a softmax layer was used in the output to support multi-class classification. The dataset was divided into two parts: 80% for training and 20% for testing. All models were trained using training data. A 5-fold cross-validation was conducted to assess the models performance and ensure that the results were consistent across various data subsets.

4) Evaluation Model
The performance of each model was assessed using accuracy, precision, recall, and F1-score for all outcome categories (win, draw, and loss). In addition, the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) was measured to evaluate how well the models could differentiate between the classes. The confusion matrix was analyzed to identify misclassification patterns, particularly for the underrepresented drawn class.

IV. RESULT AND DISCUSSION

The dataset was divided into separate subsets for training and testing, with particular attention given to the chronological structure and timing patterns typical of football seasons. To ensure a robust evaluation of the model performance, the test set was designed to encompass all matches from a single season, specifically the 2023/2024 season, comprising 380 games. This approach accounts for the variability in team performance across a season, where factors such as initial adaptation at the season's outset, fatigue, or the attainment of seasonal objectives toward the conclusion may lead to atypical outcomes. Such fluctuations can result in unexpected match results, necessitating a comprehensive test set spanning an entire season to validate the credibility of the classification model used.

To construct the training set, data from four preceding seasons (2023/2024) were utilized, totaling 380 matches. This extensive training corpus was selected to mitigate the risk of overfitting, which may occur when models are trained on limited data, leading to excessive adaptation to the training set and a diminished predictive generalization. By incorporating a substantial number of matches across multiple seasons, the training set was designed to capture a broad spectrum of team and match dynamics, thereby enhancing the robustness of the model.

To identify the optimal classification model, a range of algorithms with distinct methodological characteristics were evaluated for their suitability for the dataset. The subsequent sections detail the specific algorithms employed and the corresponding R software libraries utilized for their implementation. Subsequently, the following algorithms were employed along with their corresponding R libraries: Naive Bayes via the e1071 package, K-nearest neighbors implemented through kknn, Random Forest using randomForest, Support Vector Machines via the svm function in e1071, C5.0 decision trees with C50, XGBoost via xgboost, Multinomial Logistic Regression using the multinom function in nnet, and Artificial Neural Networks through the nnet function of the nnet package.

Prior to evaluating the classification algorithms, the dataset was normalized using the z-score standardization method to mitigate the impact of substantial variations in feature scales, as described in. This normalization ensured that all variables contributed equitably to the predictive models, thereby enhancing their stability and performance of the models. Following normalization, a feature selection process was undertaken to identify the most salient variables for predicting match outcomes, thereby optimizing the forecasting models by focusing exclusively on the most informative features.

To determine the most relevant features, the Boruta algorithm, a heuristic feature selection method based on the random forest framework, was employed. The algorithm systematically evaluates the importance of each variable by comparing its predictive utility against randomized shadow features, aiming to retain only those variables that exhibit statistically significant relevance. This approach is advantageous because it avoids suboptimal solutions, striving instead to identify all variables containing meaningful information for prediction. The application of the Boruta algorithm resulted in the elimination of seven non-relevant variables, yielding a refined set of 18 predictive features for subsequent use in constructing the classification models.

A. Prediction Results Imbalanced Data - Phase 1

To assess the performance disparities among the classification models, multiple evaluation metrics were employed, covering the overall model accuracy and rate of correctly forecasted outcomes for draws, home team wins, and away team wins. Additionally, given the intended application of the forecasting model within a predictive decision-support system, the outcome of the predictions was evaluated by displaying a confusion matrix.

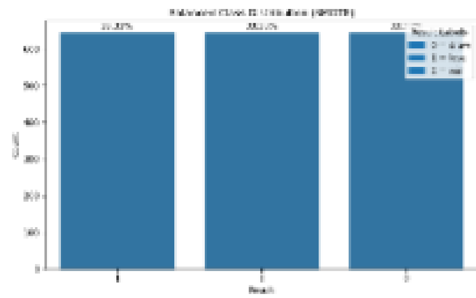

Fig. 1. Without SMOTE

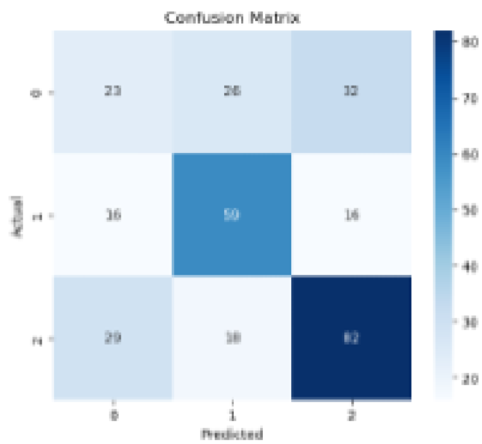

Fig. 2. With SMOTE

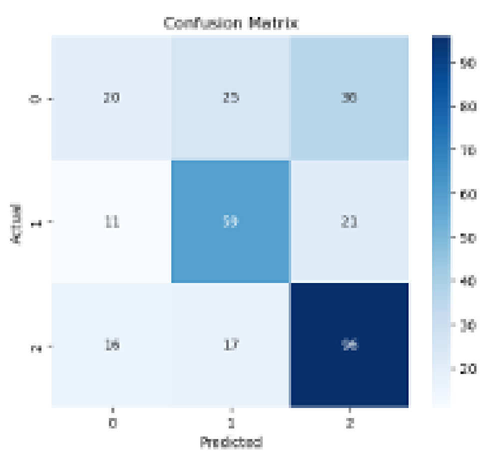

Fig. 3. Imbalanced Random Forest Confusion Matrix

B. *Prediction Results Balanced Data – Phase 2*

The provided confusion matrices illustrate the performance of a predictive model for football match outcomes, categorizing the results into win, lose, and draw. Across the

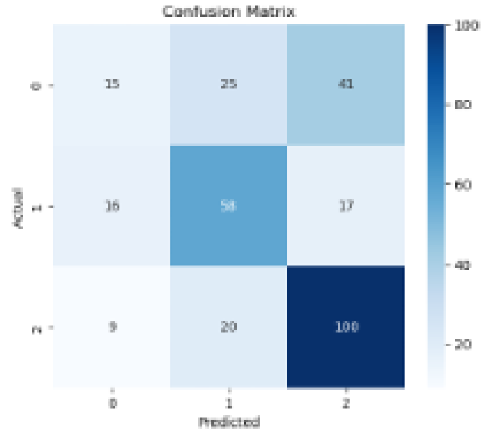

Fig. 4. Imbalanced XGBoost Confusion Matrix

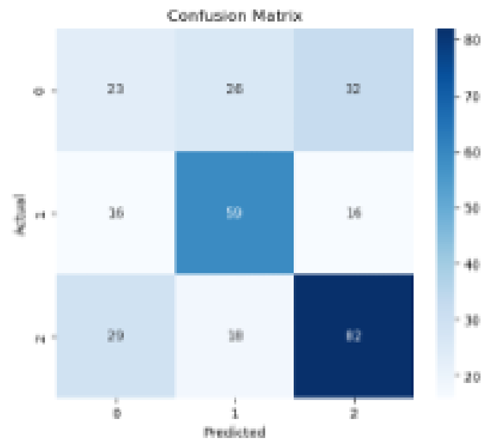

Fig. 5. Imbalanced AdaBoost Confusion Matrix

TABLE I
IMBALANCED DATA MODEL ACCURACY

Model	Accuracy
Random Forest	54%
XGBoost	58%
AdaBoost	57%

three matrices in predicting wins, the highest concentration of correct predictions (82, 93, and 103) was along the diagonal, where actual wins aligned with predicted wins. Predictions for losses showed moderate success, peaking at 59, 58, and 58 correct classifications, indicating a reasonable ability to identify losing outcomes. However, the model struggled more with predicting draws, with lower correct predictions (21, 28, and 11) and notable misclassifications, such as draws being predicted as wins or losses (e.g., 32, 42, and 42 instances). matrices reveal a trend of improving win predictions over time, while draw predictions remain the weakest, suggesting that the model is better tuned to distinguish decisive outcomes than balanced ones. Overall, the data highlight the model's strength in identifying winning scenarios, with room for enhancement in accurately forecasting results., The model consistently demonstrates strong accuracy.

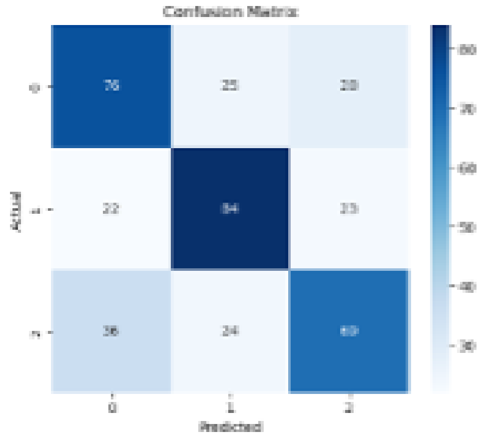

Fig. 6. Balanced Random Forest Confusion Matrix

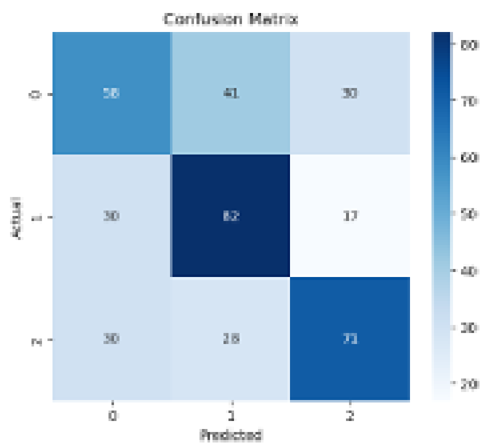

Fig. 7. Balanced XGBoost Confusion Matrix

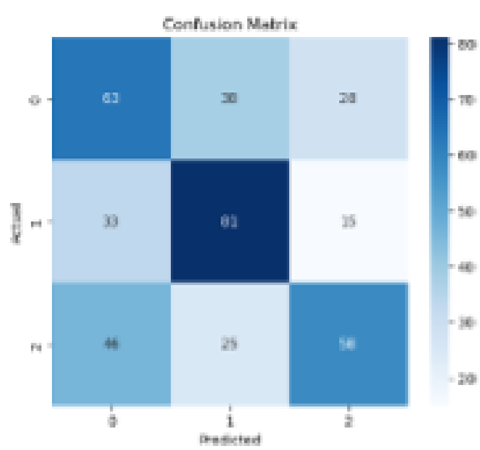

Fig. 8. balanced AdaBoost Confusion Matrix

TABLE II
BALANCED DATA MODEL ACCURACY

Model	Accuracy
Random Forest	59%
XGBoost	55%
AdaBoost	52%

We ran AdaBoost, Random Forest, and XGBoost prediction models and implemented SMOTE on each model. From the experiment, we found that Random Forest balanced data using SMOTE produced the best results on every condition, that is, away, home, or draw, with the highest accuracy of 59%, precision of 59%, recall of 59%, and F1-Score of 59%.Continued with the highest ROC AUC of 61% on Random Forest Balanced Data.With these results, we need to find out more about the contribution of each attribute using SMOTE, which allows an explanation of the contribution of each attribute in making predictions.

TABLE III
BALANCED DATA MODEL ACCURACY

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Random Forest	59%	59%	59%	59%	61%
XGBoost	55%	54%	54%	54%	59%
AdaBoost	52%	53%	52%	52%	58.4%

V. CONCLUSIONS

In this study, we outline the design and implementation of machine learning models intended to predict football match outcomes, providing support for sports betting applications. We drew on data from two distinct sources: one providing historical match statistics and the other supplying team-related information. By conducting a thorough data analysis and preprocessing, we derived essential insights into the variables that were most critical to our predictive models. This study involved comparing various algorithms to identify the most effective one for match outcome prediction. These algorithms were trained on data spanning 5 leagues with 380 matches on the 2023/2024 football season evaluated using all matches from the 2023/2024 season. This approach allowed us to closely examine the model's performance across seasons, focusing on metrics such as the accuracy of match predictions and the profit margins achievable in weekly betting scenarios. Our final model achieved a prediction accuracy of 59%, a performance that rivals the best studies in this domain. Moreover, the profit margin outperformed that reported in the referenced case studies. Moving forward, we plan to incorporate this predictive model into a decision support system that will evaluate the risk associated with coaching decisions based on the model probability outputs. This system will provide readers and especially teams and coaches with clear insights into the leagues, teams, players environment to implement their own methods and systems towards their own team .

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